Rings: Units, Zero Divisors, Idempotents, Nilpotents, Integral Domains, Fields, Characteristic

From the class book: Chapter 14, Sections 14.1 to 14.8

1. Key definitions

Ring. $(R, +, \cdot)$ with $(R, +)$ an abelian group, multiplication associative, and both distributive laws. R is commutative if $ab = ba$; it has unity if some 1 satisfies $1a = a1 = a$ (unique when it exists). CRU = commutative ring with unity.

Zero divisor. A non-zero a with $ab = 0$ or $ba = 0$ for some $b \neq 0$.

Unit. In a ring with unity, a is a unit if $ab = ba = 1$ for some b. Units form the group $U(R) = R^{\times}$. (Unity is one particular unit; -1 is a unit of $\mathbb{Z}$ that is not the unity.)

Integral domain. A CRU with $1 \neq 0$ and no zero divisors; equivalently cancellation $ab = ac,\ a \neq 0 \Rightarrow b = c$ holds.

Division ring; field. A ring with $1 \neq 0$ in which every non-zero element is a unit; a field is a commutative division ring.

Characteristic. The least $n \geq 1$ with $nx = 0$ for all $x \in R$; $\mathbf{char}\, R = 0$ if none exists. With unity, $\mathbf{char}\, R$ is the additive order of 1.

Idempotent, nilpotent, unipotent. a is idempotent if $a^2 = a$; nilpotent if $a^n = 0$ for some n (least such n = index); unipotent if $1 - a$ is nilpotent. A ring is Boolean if $x^2 = x$ for all x.

2. Main results

Theorem (commutativity criteria)
(i) R is commutative iff $(a+b)^2 = a^2 + 2ab + b^2$ for all a, b iff $(a+b)(a-b) = a^2 - b^2$ for all a, b. (ii) If R has unity and $(xy)^2 = x^2y^2$ for all x, y, then R is commutative (substitute $y + 1$, then $x + 1$). (iii) $x^2 = x$ for all x, or $x^3 = x$ for all x, forces commutativity. (iv) If $(R, +)$ is cyclic, R is commutative; so every ring of prime or square-free order is commutative. (v) Every ring with unity of order p^2 is commutative.

The unity in (ii) cannot be dropped: $\left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{Z}_2 \right\}$ satisfies $(xy)^2 = x^2y^2$ but is non-commutative. It is the smallest non-commutative ring (order 4, no unity); the smallest non-commutative ring with unity is $T_2(\mathbb{Z}_2)$, order 8.

Theorem (direct sums)
$R_1 \oplus \cdots \oplus R_n$ is commutative (has unity) iff each R_i is; $U(R_1 \oplus \cdots \oplus R_n) = U(R_1) \times \cdots \times U(R_n)$; units, idempotents and nilpotents of a product are tuples of such elements. A product of two non-zero rings has zero divisors $(1, 0)(0, 1) = 0$, so is never an integral domain.

Theorem (the ring Z_n)

A non-zero $\bar{a} \in \mathbb{Z}_n$ is a unit iff $\gcd(a, n) = 1$ and a zero divisor otherwise. Hence $|U(\mathbb{Z}_n)| = \varphi(n)$, the number of zero divisors is $n - \varphi(n) - 1$, and $\mathbb{Z}_n$ is an integral domain iff it is a field iff n is prime. If $n = p_1^{a_1} \cdots p_k^{a_k}$: $\mathbb{Z}_n$ has 2^k idempotents and $n/(p_1 \cdots p_k)$ nilpotents (the multiples of $p_1 \cdots p_k$).

Theorem (finite rings)

A finite integral domain is a field ($x \mapsto ax$ is injective, hence onto, so $ax = 1$). In a finite ring with unity every non-zero element is a unit or a zero divisor. (Wedderburn) Every finite division ring is a field.

Theorem (characteristic)

The characteristic of an integral domain is 0 or a prime (if $n = rs$, then $(r \cdot 1)(s \cdot 1) = 0$). A finite integral domain (finite field) has p^n elements, $p = \mathbf{char}$. A ring of characteristic 0 is infinite; a finite ring has $\mathbf{char}\, R \mid |R|$; $\mathbf{char}(R_1 \times R_2) = \mathbf{lcm}(\mathbf{char}\, R_1, \mathbf{char}\, R_2)$, taken as 0 if either is 0.

Theorem (Freshman's dream)

In a commutative ring of prime characteristic p: $(x + y)^p = x^p + y^p$, and $(x + y)^{p^n} = x^{p^n} + y^{p^n}$. It fails for non-prime characteristic: in $\mathbb{Z}_4$, $(1 + 1)^4 = 0 \neq 2 = 1^4 + 1^4$.

Theorem (idempotents and nilpotents)

An integral domain has only idempotents $0, 1$ and only nilpotent 0. If e is idempotent, so is $1 - e$, and e, $1 - e$ are orthogonal; in a CRU, $R \cong eR \times (1 - e)R$ (e.g. $\mathbb{Z}_6 \cong 3\mathbb{Z}_6 \times 4\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$). In a commutative ring, if $a^m = 0 = b^n$ then $(a + b)^{m+n-1} = 0$, and nilpotents form an ideal. If $b^k = 0$, then $(1 - b)^{-1} = 1 + b + \cdots + b^{k-1}$: every unipotent is a unit, and unit + nilpotent = unit in a commutative ring. A non-zero nilpotent is a zero divisor and never a unit. In any ring, ab nilpotent $\Rightarrow ba$ nilpotent since $(ba)^{n+1} = b(ab)^n a$.

Theorem (Boolean rings)

If $x^2 = x$ for all x: $2x = 0$ (from $x + x = (x + x)^2 = 4x$), R is commutative, every prime ideal is maximal, a Boolean integral domain is $\mathbb{Z}_2$, and a finite Boolean ring is $\mathbb{Z}_2^n \cong \mathcal{P}(X)$, $|X| = n$, of order 2^n.

3. Standard rings and formula tables

Ring	Comm.	Unity	Int. domain	Field
$\mathbb{Z}$; $\mathbb{Z}[i]$, $\mathbb{Z}[x]$, $F[x]$	yes	yes	yes	no
$n\mathbb{Z}$, $n \geq 2$	yes	no	no	no
$\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$	yes	yes	yes	yes
$\mathbb{Z}_n$, n composite	yes	yes	no	no
$M_n(F)$, $n \geq 2$	no	yes	no	no
$\mathbb{H}$ (real quaternions)	no	yes	–	no (division ring)
$C[0, 1]$; $\mathcal{P}(X)$, $\lvert X\rvert \geq 2$	yes	yes	no	no

Groups of units

Ring	Units
$\mathbb{Z}$; $\mathbb{Z}[\sqrt{-d}]$, $d \geq 2$	$\{\pm 1\}$
$\mathbb{Z}[i]$; $\mathbb{Z}[\omega]$	$\{\pm 1, \pm i\}$ (cyclic, 4); $\{\pm 1, \pm\omega, \pm\omega^2\}$ (cyclic, 6)
$\mathbb{Z}[\sqrt{d}]$, $d > 1$ square-free	infinite, e.g. $(1+\sqrt{2})^n$; unit iff $\lvert a^2 - db^2\rvert = 1$
$F[x]$; $F[[x]]$	non-zero constants; series with $a_0 \neq 0$
$R[x]$, R commutative	$a_0 + a_1 x + \cdots$ with a_0 a unit, $a_1, \ldots, a_n$ nilpotent
$\mathbb{Z}_n[i]$	$a + bi$ with $\gcd(a^2 + b^2, n) = 1$

Norm test for $\mathbb{Z}[\sqrt{-d}]$: $N(a + b\sqrt{-d}) = a^2 + db^2$ is multiplicative; units are exactly the elements of norm 1. $\mathbb{Z}_p[i]$ is a field with p^2 elements iff $p \equiv 3 \pmod 4$; for $p = 2$ or $p \equiv 1 \pmod 4$ it has zero divisors.

Counting in Z_n

n	units $\varphi(n)$	zero divisors	idempotents	nilpotents
12	4	7	4 (0, 1, 4, 9)	2 (0, 6)
18	6	11	4	3
24	8	15	4	4
30	8	21	8	1
36	12	23	4	6
60	16	43	8	2
100	40	59	4	10

Characteristics and matrix rings over F_q

Ring	char	Ring	char
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}[x], M_n(\mathbb{Z})$	0	$\mathbb{Z}_n,\ \mathbb{Z}_n[x],\ M_k(\mathbb{Z}_n)$	n
$\mathbb{Z} \times \mathbb{Z}_n$	0	$\mathbb{Z}_m \times \mathbb{Z}_n$	$\operatorname{lcm}(m, n)$
$\mathbb{F}_{p^n}$, $\mathbb{Z}_3[i]$	p, 3	non-zero Boolean ring, $\mathcal{P}(X)$	2

$\lvert GL_n(\mathbb{F}_q)\rvert = (q^n - 1)(q^n - q)\cdots(q^n - q^{n-1})$, $\lvert SL_n\rvert = \lvert GL_n\rvert/(q-1)$, nilpotent matrices in $M_n(\mathbb{F}_q)$: q^{n^2-n}. So $M_2(\mathbb{Z}_2)$: 16 elements, 6 units, 4 nilpotents, 8 idempotents; $M_2(\mathbb{Z}_3)$: 48 units, 9 nilpotents; $M_3(\mathbb{Z}_2)$: 168 units, 64 nilpotents.

4. Exam shortcuts and standard counter-examples

> **Exam shortcut.** In $\mathbb{Z}_n$ the only element that is neither a unit nor a zero divisor is 0. In $\mathbb{Z}$, 2 is neither (infinite rings can have such elements; so can $C[0,1]$: a function with finitely many zeros, at least one).

> **Exam shortcut.** For a product, count componentwise: units $\prod \lvert U(R_i)\rvert$, idempotents $\prod$ (idempotents of R_i), nilpotents $\prod$ (nilpotents of R_i). $\operatorname{char}(\mathbb{Z}_m \times \mathbb{Z}_n) = \operatorname{lcm}(m, n)$, not mn.

Standard counter-examples. $2\mathbb{Z}$ has no unity. The subring $\{0,2,4,6,8\} \subset \mathbb{Z}_{10}$ is a field with unity 6, not 1. $\left\{\begin{pmatrix} a & a \\ a & a \end{pmatrix}\right\}$ is a field isomorphic to $\mathbb{R}$ with unity $\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$. $M_2(\mathbb{R})$ with entrywise product is commutative with unity but has zero divisors. Quaternions over $\mathbb{Z}_3$: $(1+i+j)(1-i-j) = 1+1+1 = 0$, not a division ring. In $\mathcal{P}(X)$ every non-empty proper subset is a zero divisor. In a non-commutative ring the sum of nilpotents need not be nilpotent: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ squares to I.

Exam shortcut. A finite integral domain is a field, but an infinite one need not be ($\mathbb{Z}$, $F[x]$). Group rings RG with $|G| > 1$ finite always have zero divisors: $(1-g)(1+g+\cdots+g^{m-1}) = 1-g^m = 0$.

5. Solved examples

Example 1. Find the number of units, zero divisors, idempotents and nilpotents of $\mathbb{Z}_{72}$.

Solution. $72 = 2^3 \cdot 3^2$. Units: $\varphi(72) = 72 \cdot \frac{1}{2} \cdot \frac{2}{3} = 24$. Zero divisors: $72 - 24 - 1 = 47$. Idempotents: $2^2 = 4$. Nilpotents: multiples of $\operatorname{rad}(72) = 6$, so $72/6 = 12$.

Example 2. In $\mathbb{Z}_4 \times \mathbb{Z}_{12}$, count units, idempotents and nilpotents, and find the characteristic.

Solution. Units: $\varphi(4)\varphi(12) = 2 \cdot 4 = 8$. Idempotents: $\mathbb{Z}_4$ has 2 ($0, 1$), $\mathbb{Z}_{12}$ has 4, giving 8. Nilpotents: $\mathbb{Z}_4$ has $\{0,2\}$, $\mathbb{Z}_{12}$ has $\{0,6\}$, giving 4. $\operatorname{char} = \operatorname{lcm}(4,12) = 12$.

Example 3. How many units does $\mathbb{Z}_{13}[i]$ have? Is it an integral domain?

Solution. $a+bi$ is a non-unit iff $a^2+b^2 \equiv 0 \pmod{13}$. Since $5^2 = 25 \equiv -1$, this means $a \equiv \pm 5b$. Solutions: $(0,0)$, and for each $b \neq 0$ two values of a: $1 + 12 \cdot 2 = 25$. Units: $169 - 25 = 144$. Not an integral domain: $(5+i)(5-i) = 26 \equiv 0$. (Indeed $\mathbb{Z}_{13}[i] \cong \mathbb{Z}_{13} \times \mathbb{Z}_{13}$, with $12^2 = 144$ units.)

Example 4. Show that $S = \{0,3,6,9,12\} \subset \mathbb{Z}_{15}$ is a field and identify it.

Solution. $S = \langle 3 \rangle$ is closed under subtraction and multiplication. $6 \cdot 3 = 18 \equiv 3$, $6 \cdot 6 = 36 \equiv 6$, $6 \cdot 9 = 54 \equiv 9$, $6 \cdot 12 = 72 \equiv 12$: so 6 is the unity of S. Inverses: $3 \cdot 12 = 36 \equiv 6$, $9 \cdot 9 = 81 \equiv 6$, $6 \cdot 6 \equiv 6$. Every non-zero element is a unit and S is commutative, so S is a field of order 5, $S \cong \mathbb{Z}_5$ (via $6 \mapsto 1$). Its unity differs from that of $\mathbb{Z}_{15}$.

Example 5. Which are units in $\mathbb{Z}_8[x]$: (a) $3+4x+2x^3$, (b) $2+x$, (c) $5+6x^2$?

Solution. A polynomial over a commutative ring is a unit iff its constant term is a unit and other coefficients are nilpotent. In $\mathbb{Z}_8$, units are $1,3,5,7$ and nilpotents are $0,2,4,6$. (a) unit (3 unit; $4, 2$ nilpotent). (b) not a unit (2 is not a unit). (c) unit. For instance $(1+4x)^2 = 1+8x+16x^2 = 1$.

Example 6. Let R be a ring with $x^3 = x$ for all x. Show that $6x = 0$, and decide whether $\operatorname{char} R$ must be 6.

Solution. $(2x)^3 = 2x$ gives $8x = 2x$, so $6x = 0$. Also R has no non-zero nilpotent (if $x^n = 0$ then $x = x^3 = x^5 = \cdots = 0$). The characteristic divides 6 but need not equal 6: $\mathbb{Z}_2$ and $\mathbb{Z}_3$ satisfy $x^3 = x$ with characteristics 2 and 3; $\mathbb{Z}_6$ has characteristic 6.

Example 7. Count the units of the ring $T_2(\mathbb{Z}_3)$ of upper triangular 2×2 matrices, and compute $(2+3)^5 - 2^5 - 3^5$ in $\mathbb{Z}_5$.

Solution. $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ is invertible iff $ac \neq 0$: $2 \cdot 2 \cdot 3 = 12$ units. In $\mathbb{Z}_5$ (characteristic 5), Freshman's dream gives $(2+3)^5 = 2^5 + 3^5$, so the answer is 0.